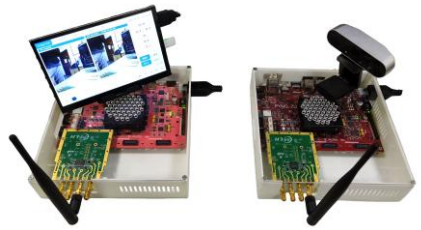
GlideLink: Real-Time DeepJSCC-Q Image Transmission on AMD FPGA

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On-board test: RX (left), TX (right)
AMD Zynq UltraScale+ MPSoC

INTRODUCTION

Live video from drones, robots and remote cameras must cross radio links whose SNR changes from moment to moment. **Separate source-channel coding (SSCC)** compresses the image with JPEG and adds a channel code: below the code threshold, bit errors propagate through the variable-length code and the whole image is lost (the **cliff effect**); above it, quality no longer improves.

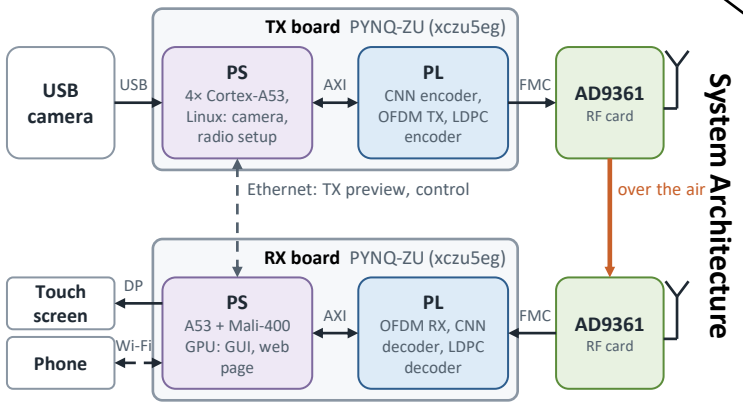
Typical application case



Deep joint source-channel coding (DeepJSCC) maps pixels directly to channel symbols with a pair of CNNs, so image quality follows the SNR smoothly. **DeepJSCC-Q** (Tung et al., JSAIT 2022) restricts the symbols to a QAM constellation that a standard digital PHY can carry, but it is computation-heavy and the receiver must pass undecided soft symbols to the decoder.

GlideLink builds a complete real-time link on two AMD Zynq UltraScale+ MPSoC boards (PYNQ-ZU) with AD9361 radios. The DeepJSCC-Q encoder, the decoder and an 802.11a-like OFDM PHY all run in the PL and send **256 × 256 RGB images at 30 fps over the air**. The PS captures camera frames, sets up the radios and shows the live link on a touch screen.

Key technical issues: fitting a 1.15-GMAC-per-frame CNN codec into one FPGA at 30 fps with all weights on chip; synchronizing and tracking 683-symbol OFDM frames precisely enough for soft-symbol decoding; and a fair over-the-air comparison with JPEG + LDPC on the same PHY.



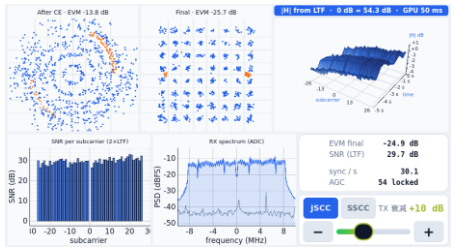
Key innovations and technical issues addressed:

1. Streaming CNN accelerator: W8A12 integer layers on 64 convolution engines in a layer pipeline at 250 MHz; all weights and activations stay on chip, no DDR traffic.

2. Asymmetric shallow decoder: trained jointly with the encoder, it needs 1/102 of the original multiply-accumulates. The codec does 5.2× the workload of a published FPGA DeepJSCC (GLOBECOM 2025) with fewer DSPs and BRAMs.

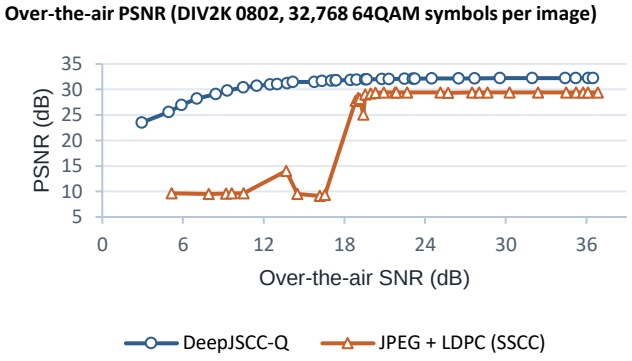
3. Soft-symbol OFDM PHY: 683-symbol frames with PN scrambling and second-order SFO/CPE tracking; equalized I/Q go to the decoder without hard decisions.

4. Live system: camera to touch screen with live PSNR, constellations and a 3-D channel view on the Mali-400 GPU.

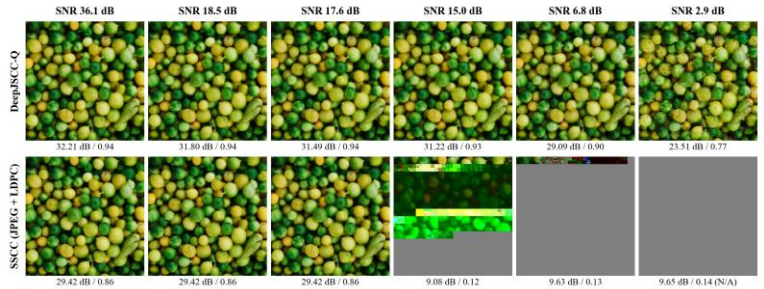


CREATIVE DESIGN

RESULT



DeepJSCC-Q is **2.8 dB** above JPEG + LDPC at high SNR and still gives **23.5 dB at 2.9 dB SNR**, while the baseline fails abruptly at 16 to 19.5 dB.



30 fps for 10 min (18,020 frames), **67.5 ms** PL end-to-end latency

Design	LUT	BRAM36	DSP	WNS (ns)	Power*
TX	54,241	104.5	243	+0.082	3.07 W
RX	63,699	65.5	133	+0.307	3.44 W

* measured, PS + PL + DDR; xczu5eg after routing